

Event-based PIV for Flow Visualization in Laser Ablation Processes

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ABSTRACT

Laser-based manufacturing processes typically require fluid flow for fume exhaustion, inert chemical interaction, cooling, etc. Traditionally, measuring flow field characteristic requires high expertise and expensive high-speed equipment and therefore is not typically investigated as a parameter space. Event-based cameras have been recently developed for use as sensors for Particle Image Velocimetry (PIV) and show great results in wind tunnels and manufacturing environments as a simpler option for particle tracking. The ability to have a relatively cheap and simple method setup allows for laser process parameter and system verification.

In this work, the objective is to integrate a cost-effective, high-frequency flow-sensing method into laser machining research to better understand the role of airflow in process control. A 532nm CW laser with cylindrical optics was used to create a laser sheet to illuminate ejected debris from laser ablation processes. Multiple materials, fluences, hatch patterns, and bulk laser power ablation experiments were performed on our 300W Amplitude Tangor workstation while measuring particle and fume extraction of the flow field. These in-situ measurements gave insight into the volatility of the laser ablation process and the best practices for predictable laser manufacturing. As ultrashort pulsed industrial laser processing emerges on large areas and complex surfaces, developing methods to retain expected laser-material interactions becomes very important.

Keywords: PIV, event-based camera, ablation, ultra-short pulse lasers, USPL, fume extraction, large area processing, fluid flow

1. INTRODUCTION

The average power of commercially available ultrashort pulsed lasers (USPL) has increased significantly in the last two decades. From Amplitude's first laser being a 1W system in 2001, to releasing a 300W USPL laser in 2019 [1]. This significant increase in USPL laser power, allows for machining or modifying large surface areas at a rate that is viable for commercial manufacturing applications. The variability of fume extraction on the quality of USPL machining has been largely overlooked due to the volumetric removal rate (mm^3/s) being limited with lower USPL average powers. In addition, USPL ablation can create harmful nanoparticles during the ablation process [2]. When machining large areas in manufacturing environments, the system designers and engineers need to take great effort in verifying that these particles are being extracted and filtered to have a safe coinhabited space. For these reasons, this work was inspired to create a low cost, quick evaluation method for fume extraction design during USPL ablation processes.

Particle image velocimetry (PIV) is a very common method for evaluating fluid flow and particulate movement under dynamic pressure/temperatures. However, a traditional PIV setup is very costly, including a ND:YAG high energy nanosecond pulsed green laser, high speed camera, synchronization of laser/camera and time consuming analysis of image pairs. Event-based imaging has become commercially available in the laser decade through companies such as Prophesee.ai, iniVation, IDS, Lucid Visionlabs, Sony, etc. Event-based imaging or neuromorphic vision systems, allow for high-speed imaging at a fraction of the cost. In addition, the data rates are much lower due to the lack of frame rates which ordinarily capture an intensity at each pixel. This is due to the architecture of the sensor used. The event-based cameras only record an "event" at each pixel when there is a relative change in intensity at said pixel. This means, if a sensor was recording a wall with no light change or movement, the camera would read no recorded events. Then, if someone were to walk in front of the camera, the outline of the person would be recorded due to the increase and decrease in intensity of the pixels as seen in Figure 1 [3]. Recently, aerodynamicists have used event-based cameras to replace high-

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speed cameras in complex PIV setups. Gunasekaran et al, performed a sensor evaluation study to determine the validity of event-based cameras in predicting fluid flow and found great success, as can be seen in Figure 2 [2].

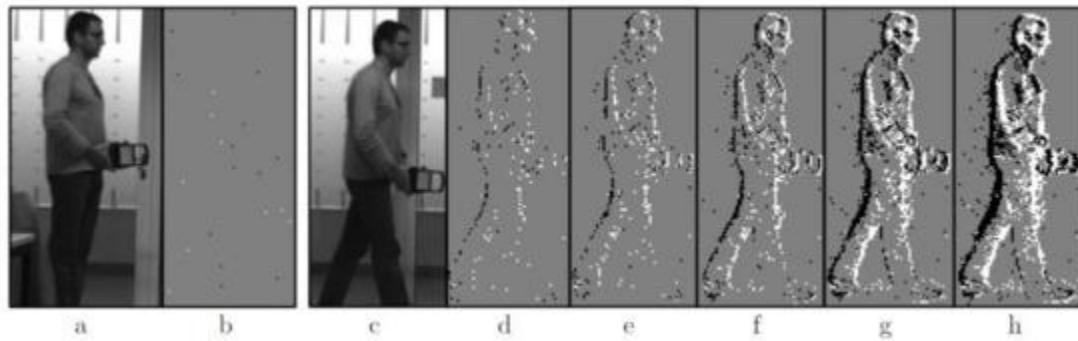


Figure 1. Silicon retina sensor in comparison to a conventional monochrome sensor. White pixels (off-events), Black pixels (on-events), Gray pixels (no events). (a) Person without movement in front of the monochrome sensor, (b) silicon retina output without movement, (c) person walking in front of the monochrome sensor, (d–h) silicon retina data from the walking person with collected events over a time period of 5 ms, 10 ms, 20 ms, 40 ms and 60 ms [3]

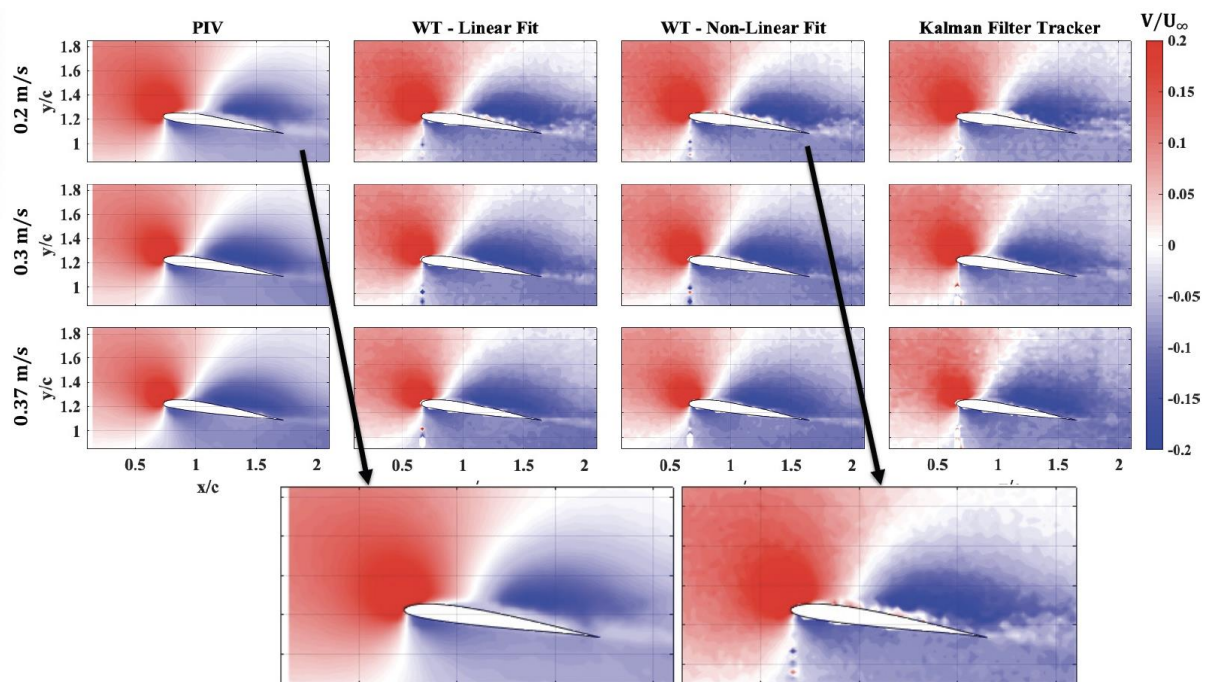


Figure 2. Event based camera PIV comparison of various algorithms.

2. EXPERIMENTAL SETUP

The development of the process was primarily done at UDRI, using their 300W Tangor, USPL laser. Various laser parameters were changed during the experiments, including average power, pulse energy, frequency, mark speed, hatch direction, interlacing, etc. The laser was rastered via an excelli-SCAN 14 with a 420mm F-theta lens. As seen in Figure 3, the event-based camera recorded a planar image irradiated by a 1W 532nm CW laser, with a laser sheet optic to highlight the ablation area parallel to the fume extraction direction. Multiple OD filters and camera lenses were trialed to obtain valuable event-based data during the experiments. The fume extraction nozzle was 3D printed at UDRI and hooked up to a BOFA AD 1000 iQ. A Prophesee EVK-14 HD camera was used to record all events. Multiple different materials were tested as well, including steel, aluminum, carbon fiber reinforced polymer (CFRP) and G10.

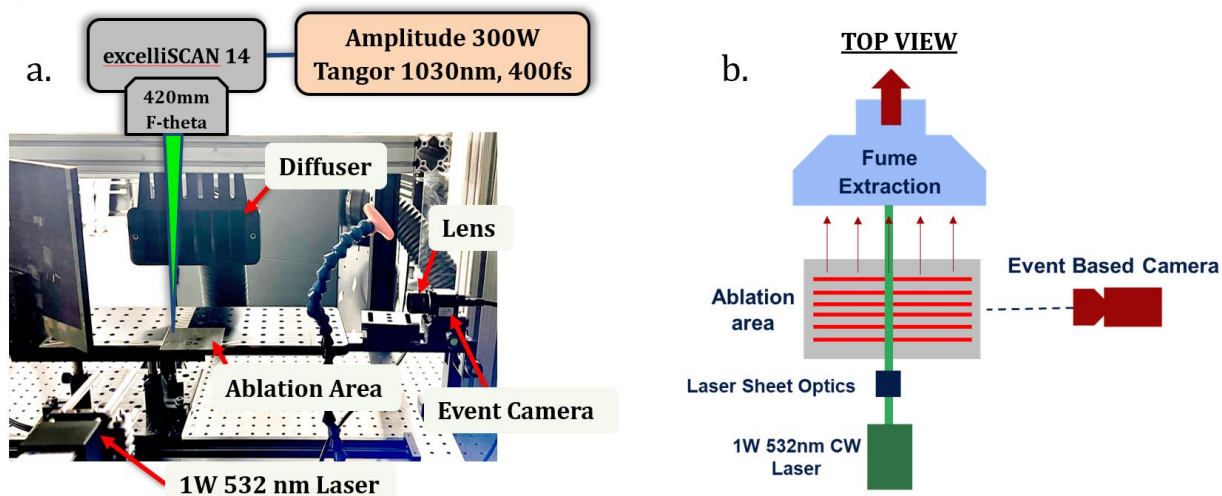


Figure 3. (a) Image of event-based PIV setup at UDRI. (b) Top-down diagram of event-based PIV setup

3. RESULTS

A few selected example cases will be shown in the results. Email the authors for more information or to obtain the actual videos. As shown in Figure 4, this is an example frame of the event-based camera while performing laser ablation experiments.

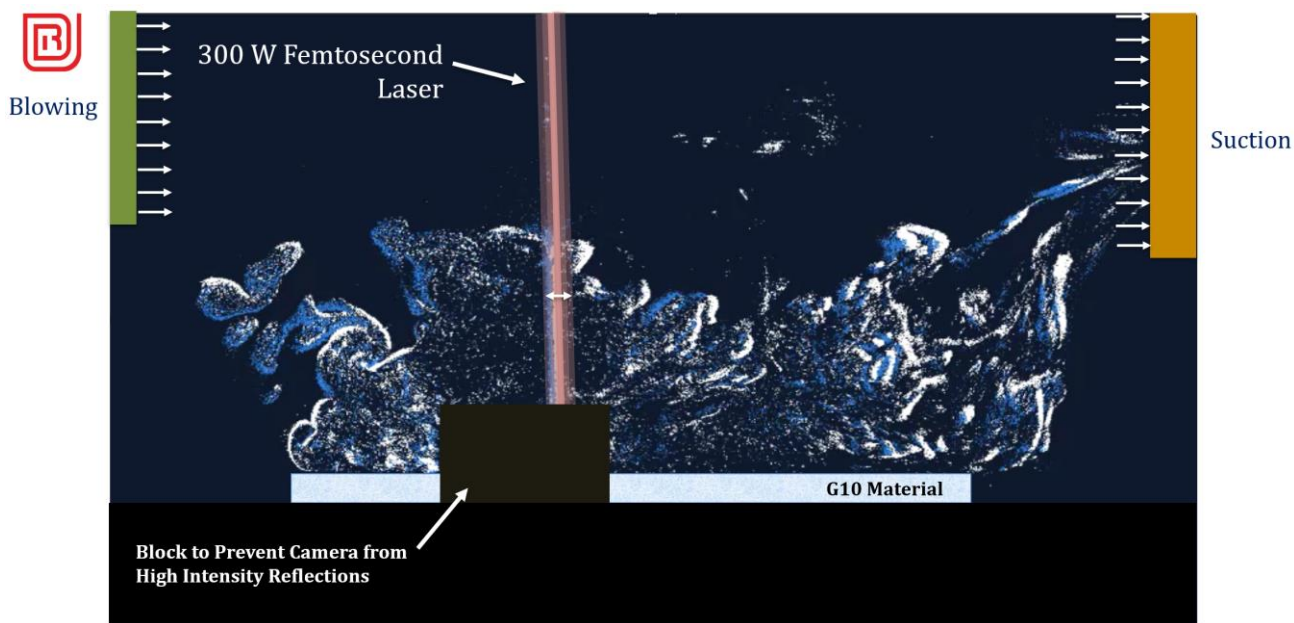


Figure 4. Example of event-based camera frame during laser ablation

Fume Extraction Volumetric Removal

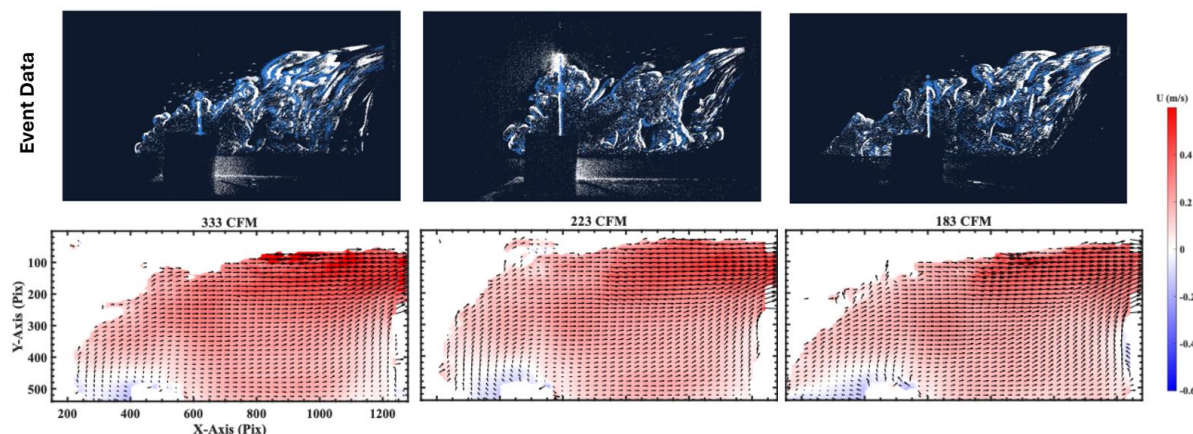


Figure 5. A comparison of BOFA fume extraction settings is shown at 333, 223 and 183 CFM. The top row is example images as seen in the by the event-based camera software and the bottom row are U velocity plots generated by a clustering-particle tracking PIV approach.

As seen in Figure 5, the CFM was changed while retaining the same laser processing parameters. In the images provided, the 183 CFM saw the largest area of events and many more events occurring on the left side of the ablation area (further from fume extraction). This indicates that there is still some particulate not being extracted during the ablation process.

Hatch Direction

Hatch Direction – Fiberglass Composite G10

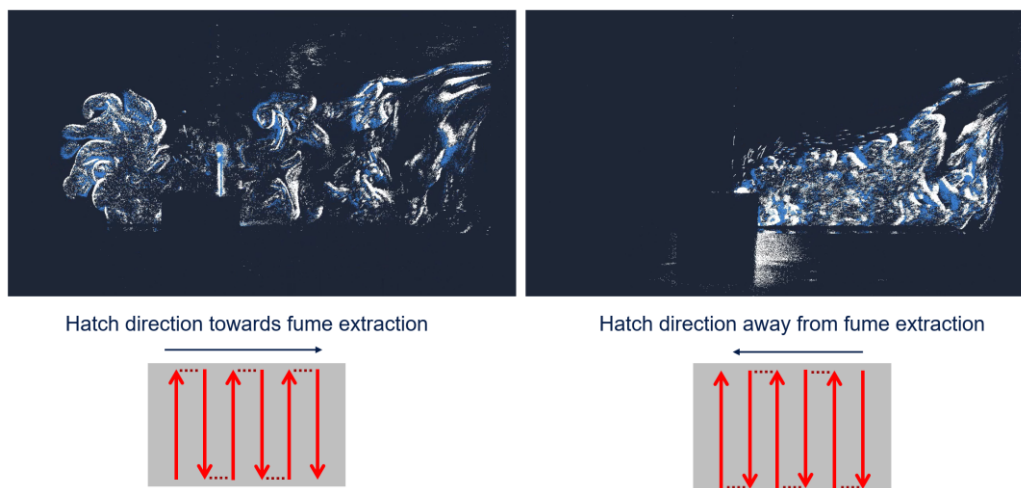


Figure 6. Example images of laser hatch direction during fume extraction

Hatch direction is a very common laser parameter that mainly gets heavily evaluated in laser powder bed fusion (LPBF) but is typically overlooked in USPL laser ablation processes except in polarization and LIPSS sensitive applications. Figure 6 shows the drastic difference in hatch direction on ablative particulate extraction. When hatching toward the fume extractor, the intense plume pressure will move to the right ahead of the debris, which creates a reaction that continuously pushes debris away from the fume extractor. Additionally, this process will increase debris-laser interference which will affect the machining quality of the manufacturing process.

CFRP Multiple Passes, Mark Speed, and Power

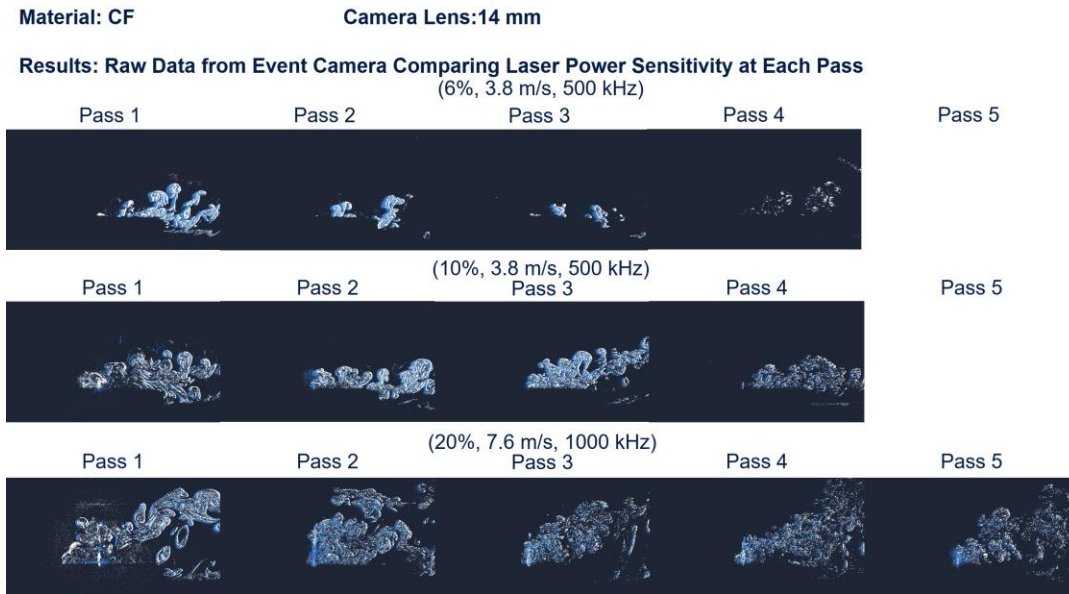


Figure 7. Parameter variability of ablation material removal rate and effective fume extraction on CFRP.

Many parameters can be evaluated quickly, even indicating the type of particulate that is being removed from a pass-by-pass basis. Figure 7 shows a snapshot of each parameter and passes from 1 to 5 on CFRP. From 15W (6%) to 60W (20%), the volumetric removal is obviously higher due to the many more events per pixel. In addition, the volumetric removal decreases with increasing number of passes. This is primarily due to the parameters being optimized for removing the epoxy resin layer and leaving the carbon fiber minimally ablated. The size of event clusters changes as passes are increased as well. This could indicate a difference between smaller organic compounds versus pieces of carbon fiber being removed during the laser ablation process. Additionally, Figure 8 shows an example of the velocity magnitudes for the same processing parameters. Therefore, in the first three passes majority of the resin is removed, revealing a much higher event area. This example gives strong indication for using event-based cameras to determine processing parameters for multi-material stack-ups.

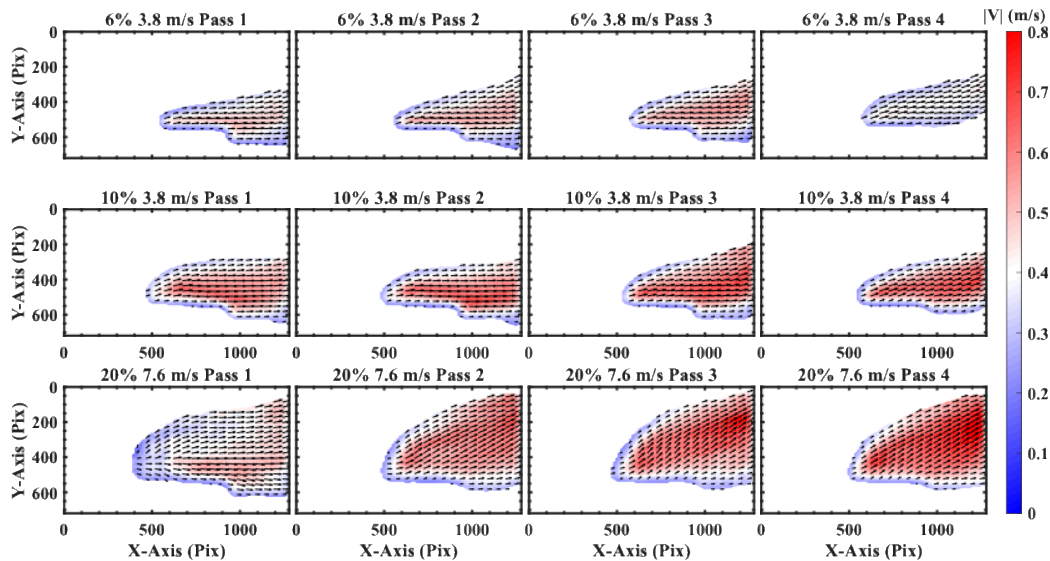


Figure 8. Velocity Magnitude of Laser Ablation Processing on CFRP.

4. CONCLUSIONS

In conclusion, experimentation methods were successful at creating an economic PIV setup for laser ablation. The reduced cost of the laser, camera and system setup could save companies \$100k or more and reduce the size of PIV setup for manufacturing applications. The event-based cameras recorded differences in many laser parameters and composite material characteristics. The PIV algorithms developed were successful at creating vector graphs indicative of adequate fume extraction and optimized laser parameters. A hybrid clustering/particle tracking approach was developed to handle changes in particulate size/grouping, this will be discussed later and is not including in these conference proceedings.

Next Steps

Much more work still needs to be done to get the most out of this measurement technique for laser processing applications. Next steps for UDRI include machining larger areas and higher power (>150W), analyzing fume extraction during robotic processing on complex surfaces, apply to other applications such as welding and LPBF, and use AI/ML to predict laser parameters and materials from event-based data.

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